Amlan Jyoti Sarmah

+91-9395169676 | amlan.j.sarmah@gmail.com | linkedin.com/in/amlan-jyoti-sarmah |
github.com/AmlanJSarmah
Bhopal, Madhya Pradesh

Summary
Full-stack developer with a strong interest in machine learning, experienced in building production-ready web applications using React, Node.js, TypeScript, and Python. Skilled in designing REST APIs, working with SQL and NoSQL databases, and implementing ML models using Scikit-learn and PyTorch. Also experienced in low-level C++ systems programming. Thrives in projects that require both engineering rigour and practical problem solving.

Education
- B.Tech in Computer Science and Engineering, VIT Bhopal University CGPA: 8.54

Projects

songRoom — Real-Time Music Streaming & Chat *github.com/AmlanJSarmah/songRoom*
- Built a real-time collaborative app allowing multiple users to stream songs and chat simultaneously in shared rooms.
- React.js was used on the frontend for a responsive, component-driven UI with live state updates.
- Node.js, Express, and Socket.io powered the backend and WebSocket layer for synchronised playback and low-latency chat.
- MongoDB was used to store user data, room metadata, and song history.

Smart Tutor — Adaptive Learning System for Primary Schools *Demo | GitHub*
- Built an adaptive tutoring system that adjusts question difficulty in real time based on individual student performance.
- Scikit-learn was used to train a performance classifier; Flask deployed it as a REST API for difficulty predictions.
- TypeScript, Express, and Zod handled authentication, question recommendation, and request validation in a separate backend service.
- MongoDB stores student profiles, performance records, and question banks to support the recommendation engine.

Custom C++ ML Library *ml-from-scratch-cpp.vercel.app | GitHub*
- Implemented a lightweight, dependency-free ML library in C++ covering Linear Regression, Logistic Regression, and Naive Bayes with a modular class-based architecture.
- A TypeScript API exposes model training and inference endpoints, interfacing directly with the custom C++ models.
- A Python benchmark script compares performance against scikit-learn; results are visualised on a React frontend deployed on Vercel.

Leadership Experience

Technical Team Lead, North East Club, VIT Bhopal *Aug 2023 to Present*
- Led the technical team, overseeing development decisions to grow the club's digital presence.
- React.js and Tailwind CSS were used to build the club's official responsive website, improving student engagement and information accessibility.
- JavaScript and REST APIs were integrated for dynamic content updates, streamlining event announcements and reducing manual management overhead.
- Git and GitHub were used for version control, enabling smooth collaboration across a team of designers and developers for 5+ club events.

Skills
- **Languages:** JavaScript, TypeScript, C++, Python
- **Frontend:** React.js, Next.js, Tailwind CSS, Socket.io
- **Backend:** Node.js, Express.js, Flask, REST APIs, Zod, WebSockets
- **Databases:** SQL (PostgreSQL), NoSQL (MongoDB)
- **ML & Data:** Scikit-learn, PyTorch, Pandas, NumPy
- **Tools:** Git, GitHub, Vercel, Chrome Extension APIs (Manifest V3)